🚀 CP Bootcamp 2026 — Day 9

Theme: Two Pointer + Sliding Window Foundation

> **Main Goal:**
> আজ থেকে তুমি একটা বড় optimization idea শিখবে।

এতদিন:

```
Nested Loop
+
Check All Possibilities
```

করেছো।

এখন:

```
Observe Pattern
↓
Avoid Repeated Work
↓
Move Pointers / Window
↓
O(n) Solution
```

📖 Day 9 Chapter Map

```
Day 9
│
├── Chapter 1 — Why Optimization Patterns Matter
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├── Chapter 2 — Two Pointer Mental Model
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├── Chapter 3 — Two Pointer on Sorted Arrays
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```

Chapter 1 — Why Optimization Patterns Matter

ধরো problem:

> Array-তে দুইটি number-এর sum target কিনা বের করো।

Array:

```
2 4 6 8 10
```

Target:

```
12
```

Beginner Approach

সব pair check:

```
for(i=0;i<n;i++)
{
    for(j=i+1;j<n;j++)
    {
        if(arr[i]+arr[j]==target)
        {
            found=1;
        }
    }
}
```

Complexity:

```
O(n²)
```

যদি:

```
n = 100000
```

তাহলে:

```
10,000,000,000
```

বার check!

Impossible.

Optimization idea:

```
Can we use order?
```

যদি array sorted হয়:

```
2 4 6 8 10
```

তাহলে:

```
left                right

 2  4  6  8  10
 ↑              ↑
```

এখান থেকেই:

Two Pointer

Chapter 2 — Two Pointer Mental Model

Two Pointer মানে:

> একই data structure-এর উপর দুইটি pointer দিয়ে কাজ করা।

Example:

Array:

```
1 3 5 7 9
```

Pointers:

```
left → শুরু

right → শেষ
```

Visualization:

```
left
 ↓
1 3 5 7 9
        ↑
       right
```

Pointers move করে:

```
left++

right--
```

কেন Two Pointer?

কারণ:

একই element বারবার check না করে:

```
Information ব্যবহার করে
Pointer move করা
```

Chapter 3 — Two Pointer on Sorted Arrays

Problem

Sorted array:

```
1 2 4 7 9
```

Target:

```
10
```

Find pair।

Start:

```
left = 1
```

```
right = 9
```

Sum:

```
1+9 = 10
```

Found।

আরেক example:

Array:

```
1 3 5 8 12
```

Target:

```
13
```

Step 1:

```
1 + 12 = 13
```

Found।

যখন Sum ছোট

Example:

```
1 + 8 = 9
```

```
target = 13
```

Need bigger number।

তাই:

```
left++
```

যখন Sum বড়

Example:

```
5 + 12 = 17

target = 13
```

Need smaller number।

তাই:

```
right--
```

Decision Rule

```
sum < target

→ left++


sum > target

→ right--


sum == target

→ answer
```

Implementation

```
int twoSum(int arr[], int n, int target)
{
    int left = 0;
    int right = n-1;


    while(left < right)
    {
        int sum = arr[left] + arr[right];


        if(sum == target)
        {
            return 1;
        }

        else if(sum < target)
        {
            left++;
        }

        else
        {
            right--;
        }
    }


    return 0;
}
```

Complexity:

```
O(n)
```

Chapter 4 — Opposite Direction Two Pointer

এই pattern:

```
Start
 ↓

←       →

End
```

Common uses:

1. Pair Sum

```
left + right
```

2. Reverse Array

Example:

Before:

```
1 2 3 4 5
```

Swap:

```
1 ↔ 5

2 ↔ 4
```

After:

```
5 4 3 2 1
```

Code:

```
void reverse(int arr[], int n)
{
    int left=0;
    int right=n-1;


    while(left<right)
    {
        int temp=arr[left];

        arr[left]=arr[right];

        arr[right]=temp;


        left++;

        right--;
    }
}
```

3. Palindrome Check

String:

```
madam
```

Compare:

```
m == m

a == a

d == d
```

Pattern:

```
Two Pointer

+

Comparison
```

Chapter 5 — Same Direction Two Pointer

এখানে দুই pointer একই দিকে যায়।

Example:

```
slow →
fast →
```

Common use:

- Remove duplicates
- Move zeros
- Filtering

Example — Remove Duplicates

Sorted array:

```
1 1 2 2 3
```

Need:

```
1 2 3
```

Slow:

```
Unique position
```

Fast:

```
Explore
```

Visualization:

```
slow
 ↓
1 1 2 2 3
    ↑
   fast
```

Mental Model:

```
fast finds useful data

slow stores useful data
```

Chapter 6 — Sliding Window Mental Model

Sliding Window হলো Two Pointer-এর special case।

Problem:

> Continuous part (subarray/substring) নিয়ে কাজ করা।

Example:

Array:

```
1 2 3 4 5
```

Window size:

```
3
```

First window:

```
1 2 3
```

Next:

```
2 3 4
```

Next:

```
3 4 5
```

Window slide করে।

Mental Model:

```
[ window ]

Move →
Move →
Move →
```

Why Sliding Window?

Without:

প্রতিটি subarray calculate:

```
O(n²)
```

With window:

```
O(n)
```

Chapter 7 — Fixed Size Sliding Window

Problem:

> Size k subarray-এর maximum sum বের করো।

Example:

Array:

```
2 3 1 5 4
```

k:

```
3
```

Windows:

```
2 3 1 = 6
```

```
3 1 5 = 9
```

```
1 5 4 = 10
```

Answer:

```
10
```

Naive

প্রতিবার sum:

```
O(n*k)
```

Sliding Window

First sum:

```
2+3+1
```

Next:

Remove:

```
2
```

Add:

```
5
```

New:

```
3+1+5
```

Pattern:

```
Old Window

-remove outgoing

+add incoming

New Window
```

Code:

```
int maxSum(int arr[], int n, int k)
{
    int sum=0;


    for(int i=0;i<k;i++)
    {
        sum+=arr[i];
    }


    int ans=sum;


    for(int i=k;i<n;i++)
    {
        sum += arr[i];

        sum -= arr[i-k];


        if(sum>ans)
            ans=sum;
    }


    return ans;
}
```

Chapter 8 — Variable Size Sliding Window

এখানে window size fixed না।

Condition অনুযায়ী:

```
Expand
```

```
Shrink
```

Example:

> Smallest subarray whose sum >= target

Array:

```
2 3 1 2 4 3
```

Target:

```
7
```

Window:

```
2 3 1 2
```

sum:

```
8
```

Enough!

Shrink:

Remove left.

```
3 1 2
```

sum:

```
6
```

Not enough

Expand again

Mental Model:

```
Right pointer
    ↓
Expand window


Condition true?

    ↓

Left pointer
    ↓
Shrink window
```

Common Problems

Variable Sliding Window:

```
Longest substring

Smallest subarray

Maximum length segment
```

Chapter 9 — Common Bugs & Edge Cases

Bug 1 — Two Pointer without Sorting

Wrong:

```
8 2 5 1 9
```

Two sum।

কারণ movement rule কাজ করবে না।

Need:

```
Sorted Data
```

Bug 2 — Wrong Condition

Pair sum:

Wrong:

```
while(left ⩽ right)
```

Correct:

```
while(left<right)
```

কারণ:

এক element নিজে pair না।

Bug 3 — Sliding Window Remove Wrong Element

Window:

```
i-k
```

not:

```
i-k+1
```

Bug 4 — Forget Initial Window

Fixed window:

প্রথম k element আগে calculate করতে হবে।

Bug 5 — Empty Input

Handle:

```
n=0
```

Chapter 10 — Day 9 Assignment

Part A — Two Pointer Practice

Implement:

1. Reverse Array
2. Palindrome Check
3. Sorted Pair Sum
4. Count Pair Sum
5. Merge Two Sorted Arrays

Part B — Sliding Window

Implement:

1. Maximum Sum Subarray of Size K
2. Minimum Sum Subarray of Size K
3. Average of Every Window
4. Maximum Element in Every Window

Part C — Dry Run

Array:

```
2 3 1 5 4
```

k:

```
3
```

Complete:

```
Initial Window:

Sum:


Slide 1:

Remove:

Add:

New Sum:


Slide 2:

Remove:

Add:

New Sum:
```

Part D — Pattern Recognition

Problem 1

> Reverse a string

Pattern:

?

Problem 2

> Find pair with target sum in sorted array

Pattern:

?

Problem 3

> Maximum sum of k consecutive elements

Pattern:

?

Problem 4

> Longest substring without repeating character

Pattern:

?

📚 Glossary Update

Add:

```
Two Pointer

Left Pointer

Right Pointer

Slow Pointer

Fast Pointer

Opposite Direction

Same Direction

Sliding Window

Fixed Window

Variable Window

Expand Window

Shrink Window

Continuous Segment

Subarray

Substring
```

📚 Pattern Library Update

Add:

Two Pointer Pattern

Recognition:

- Sorted array
- Pair problems
- Reverse
- Palindrome

Mental Model:

left
 ↓

data

 ↑
right

Move pointers based on condition.

Sliding Window Pattern

Recognition:

- Continuous subarray
- Continuous substring
- Range problems

Mental Model:

Create Window

↓

Move Window

↓

Maintain Information

Types:

Fixed Size

Variable Size

✅ Day 9 Completion Checklist

[ ] Two Pointer idea বুঝি

[ ] Sorted array-তে pair sum solve করতে পারি

[ ] Reverse array two pointer দিয়ে করতে পারি

[ ] Palindrome check করতে পারি

[ ] Same direction pointer বুঝি

[ ] Sliding Window বুঝি

[ ] Fixed window solve করতে পারি

[ ] Variable window idea বুঝি

[ ] $O(n^2)$ থেকে $O(n)$ optimization বুঝি

[ ] কখন Two Pointer use করবো জানি

[ ] কখন Sliding Window use করবো জানি

🏁 Day 9 Final Mental Model

```
Repeated Work
        ↓
Can I remember previous information?
        ↓
Move Pointer
        ↓
Maintain Window
        ↓
O(n) Solution
```

Day 9 শেষে তোমার CP foundation:

```
Day 1-5
Pattern Recognition

+

Day 6
Functions

+

Day 7
Searching

+

Day 8
Sorting + Complexity

+

Day 9
Optimization Patterns
```

পরের **Day 10 — Recursion + Backtracking Foundation** হবে।